

#autotitrator

So this is Kong's method to measure TAN of samples in accordance to ASTM667. For use internally in MIPO. Not responsible for any misuse.

Solvent Preparation

The solvent is a mixture of 50/50/0.5 vol% toluene IPA water. Mix in bottle and shake thoroughly until it is not cloudy.

Electrode Preparation

Electrode should be stored long term in electrolyte solution. Before use, check the glass bulb for particles stuck on it. Wipe down with kimwipe until the electrode looks clear and smooth. Check the electrolyte solution (LiCl in EtOH) is filled up to the fill hole. Before use, condition in DI water for 5 minutes to hydrate glass membrane.

Titrant Preparation

Dissolve 1 to 2 small pellets of NaOH in a little bit of water, then add to 1L of IPA solution. Stir overnight. Keep it stirring when not in use to maintain solubility. If precipitate is formed (due to CO₂ in the air), filter the titrant.

Autotitrator Preparation

Ensure that the burette is filled with the titrant. If filled with DI water, flush out DI water to replace with titrant. This may require up to 10 flushes (judge until no significant phase difference in observed). If already filled with titrant, cycle the titrant burette mixture twice to ensure homogeneity.

SOP for titration

1. Complete all preparation steps mentioned above.
2. In a 250mL beaker, dissolve between 0.1g (for TAN >50) to 10g (for TAN < 5) of sample in the solvent. Measure accurately the quantity added. For calibration, use around 0.5g of KHP in ultrapure water.
3. Sonicate until completely dissolved. For MFO, some components will not dissolve properly, so stir the sample/solvent mixture for 20 minutes on a stir plate to ensure complete suspension.
4. Titrate the sample.
5. Locate the endpoint based on inflection or pH.

6. When titration is completed, rinse the probe with IPA or kimwipe soaked in the solvent solution. Then rinse with DI water and wipe down with kimwipe. When washing your beaker, soak your electrode in DI water to rehydrate the glass membrane.

Things we don't do in accordance to ASTM 667.

1. We don't replace our electrode every 6 months or wash with strongly oxidising acid solution.
2. We don't boil our IPA/NaOH solution for 3 hours in titrant prep.
3. We don't heat our samples/solvent mixture when stirring during preparation phase.